Spurious or Structural? Low-Rank Confounding and Partial Identification of Cross-Asset Price Impact

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Abstract

Cross-asset return-on-flow coefficients are routinely read as entries of a structural price-impact matrix. We show that reading is generically unidentified, characterise exactly how, and quantify what the failure costs. In a simultaneous N-asset system with K latent factors and same-bin feedback B, the gap between the population regression coefficient and the structural impact matrix has rank at most $K + \mathrm{rank}(B)$. When the structural matrix is diagonal and feedback is absent, the entire estimated cross-impact matrix is confined to the diagonal-plus-rank-K set, yet its off-diagonal entries can be as large as genuine own-impact: a strictly diagonal truth induces spurious cross terms of 0.221 against own terms spanning 0.206 to 0.395. The structural matrix is therefore set-identified rather than point-identified, and at published one-minute commonality statistics the sharp identified interval for the off-diagonal runs from -0.078 to 0.103 around an observed 0.012, so it is not identified even in sign. The same rank structure makes the consequence tractable. The execution-cost error is a rank-K quadratic form, so it vanishes on at least $N - K$ trade directions: an equal-weight index basket is mispriced by 54% of its true cost while a dollar-neutral basket is mispriced by exactly nothing, and a dollar-neutral trade has a point-identified execution cost even though the matrix behind it does not. Finally the restriction is refutable. We give a scale-free departure statistic whose rejection supports genuine cross-impact, with a bootstrap null that controls size above roughly $5N^2$ observations and is reported as invalid below. We verify the theory in a preregistered known-truth experiment with $N = 30$, $K = 3$, $T = 10^7$, and register, without reporting, a falsification protocol for the empirical stage.

1 Introduction

Consider a fact from the published cross-impact literature. Capponi and Cont [9] regress one-minute returns on the order-flow imbalance of every stock in a large cross-section and report a mean cross-asset coefficient of $+0.032$, with 23.09% of those coefficients negative. They then add a single cross-sectional principal-component control. The mean cross coefficient becomes -0.039 and the negative share rises to 84.46%.

One control flips the sign of the average estimated cross-impact. Two readings are available. Either the uncontrolled regression was contaminated and the controlled one reveals the true structural effect, or the control absorbed real impact and the uncontrolled one was right. The literature has no decisive way to choose, and the choice matters: structural impact matrices enter execution-cost models, liquidity stress tests, and manipulation constraints, where a sign error is not a rounding error.

This paper argues that neither reading is identified, and makes that claim precise rather than rhetorical.

Our starting point is the standard simultaneous system in which returns load on flows, flows load on returns within the same bin, and both load on latent common factors. The population coefficient of a return-on-flow regression equals the structural impact matrix plus a discrepancy. That much is a routine omitted-variable and simultaneity calculation, and we record it as Theorem 1 for completeness.

The paper's contribution begins one step later, with the

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structure of that discrepancy.

The confounding gap is low rank. Theorem 2 shows that the gap between the population coefficient and the structural matrix has rank at most $K + \text{rank}(B)$. The factor channel passes through a K-dimensional bottleneck and the feedback channel through the column space of B, and rank is subadditive. The bound is attained generically.

Spurious cross-impact is low rank but not small. If the structural matrix is diagonal and there is no same-bin feedback, the population coefficient matrix lies exactly in the diagonal-plus-rank-K set. This does not make the spurious off-diagonals negligible. At our registered fixture a strictly diagonal truth produces off-diagonal coefficients reaching 0.2207 while the genuine own-impact terms span 0.2061 to 0.3953. Confounding does not perturb a cross-impact matrix; it manufactures one of realistic magnitude out of nothing.

The structural matrix is set-identified. Because the gap is an unrestricted low-rank object, Proposition 2 shows that a continuum of structural matrices reproduces the same second moments. Cross-impact is therefore a partial identification problem in the sense of Manski and Tamer [25, 26, 34], a framing this literature has not used. In the permutation-invariant one-spike geometry the gap collapses to a single constant added to every entry, and Proposition 3 gives the sharp identified interval in closed form. Evaluated at published one-minute commonality statistics, that interval is $[-0.0782, 0.1027]$ around an observed off-diagonal of 0.0123: the identified half-width is 7.4 times the coefficient it is meant to pin down, and it contains zero.

The failure has a price, and a hedge. Because the cost error is the quadratic form of a rank-K matrix, it vanishes on at least $N-K$ of the N trade directions. Which directions are unsafe is determined by the confounding loadings: in the one-spike geometry an equal-weight index basket is mispriced by 54.23% of its true cost and a dollar-neutral basket by exactly nothing. A dollar-neutral trade therefore has a point-identified execution cost even though the impact matrix producing it is not identified at all. For a desk that must take factor exposure we give the closed-form schedule minimising worst-case cost over the identified set, along with the cases in which robustness provably buys nothing.

The restriction is refutable. Low rank is a testable restriction, not just a caution. Definition 1 introduces ψ_K, the relative Frobenius distance of an estimated coefficient matrix from the diagonal-plus-rank-K set. Its population value is zero under pure confounding, so a materially nonzero value is evidence *for* genuine structural cross-impact. This inverts the conventional reading, under which a large off-diagonal coefficient is itself taken as the evidence. Under our results, off-diagonal magnitude is uninformative and only departure from the low-rank set carries content.

Section 9 reports the completed known-truth verification: a preregistered $N = 30$, $K = 3$, $T = 10^7$ experiment that recovers both derived population matrices to a maximum relative discrepancy of 5.64×10^{-4} against a 10^{-3} threshold fixed in advance. Section 10 confronts the theory with published summary statistics and reports a split verdict, including the check that fails. Section 11 registers, without reporting, the falsification protocol that governs the empirical stage.

Table 1 states the evidence boundary. Only the first two rows contain completed evidence, and no claim in this paper depends on the rest.

Table 1: Research status at the manuscript freeze.

Gate	Status	Licensed statement
G0	Passed	Reproducible software and compute-plan skeleton only.
G1	Passed	Derived regression targets recover in the frozen known-truth simulation.
G2	Open	Scientific design registered; no registered benchmark, validation, or research result.
G3–G7	Locked	No market-data, identification, training, holdout, or economic claim.
G8	Locked	Final-results paper remains locked; this preprint adds no downstream result.

2 Relation to the Literature

Linear impact is the standard language for how order flow moves prices [24, 20], and underpins optimal execution [2, 29] and no-arbitrage restrictions on impact functionals [16, 32]. At the book, best-level order-flow imbalance (OFI) compresses additions, cancellations, and executions into a signed pressure statistic strongly related to short-horizon price changes [15]; multi-level OFI extends the statistic deeper [36]. Cont, Cucuringu, and Zhang [14] integrate levels by principal components and compare own-asset and cross-asset specifications in a one-minute protocol.

Cross-impact specifically has been estimated through multivariate propagator models [7] and through direct cross-sectional regressions [9]. The empirical difficulty is commonality: returns, flows, and liquidity share strong common components [21, 13, 23, 22], and Capponi and

Cont [9] argue that a common OFI component explains much of apparent contemporaneous cross-impact. Benzaquen et al. [7] state directly that individual entries of an unrestricted cross-impact matrix cannot be estimated reliably. Our contribution makes that difficulty exact: we show what the contamination looks like algebraically, prove it is low rank, and derive what remains identified.

Controlling for estimated factors is the usual remedy. Factor methods summarise large panels well [33, 10], the number of factors is estimable [4, 30, 1], and generated-regressor inference has a mature theory [3, 5]. But a predictive factor is not automatically a valid causal control, and one noisy proxy does not generically identify an unobserved confounder [27]. Theorem 1 makes that visible here, and Remark 2 shows that adding a factor control moves the estimate *along* the confounding directions rather than out of the identified set.

Our formal object – a matrix that is diagonal plus low rank – is not new as mathematics. Decompositions into sparse or diagonal and low-rank parts are well developed [12, 11, 8], and ψ_K uses a standard alternating projection. What is new here is the economic result that the confounding gap in a simultaneous impact system *must* have this structure, with rank tied to the number of latent factors and the rank of same-bin feedback, and the use of that structure as a specification test for cross-impact.

Finally, the design discipline. Broad specification searches report noise as a winner [35, 17, 31], a problem documented at scale in finance [19, 18, 6]. Preregistration is the standard response elsewhere [28] and is rare in market microstructure. Section 11 registers the empirical protocol before any data is opened.

3 Simultaneous System and Pseudo-True Coefficients

Let returns r_t, flows q_t, and idiosyncratic shocks u_t, v_t lie in $\mathbb{R}^N$, with latent factors $f_t \in \mathbb{R}^K$. Consider

$$r_t = \Lambda q_t + \Gamma f_t + u_t, \tag{1}$$
$$q_t = B r_t + \Delta_f f_t + v_t. \tag{2}$$

Here Λ is the structural contemporaneous impact matrix and B is same-bin return-to-flow feedback. All variables are centered. Assume finite second moments and a law of large numbers; mutual zero contemporaneous cross-covariances among f_t, u_t, v_t, and proxy noise; and nonsingularity of $I_N - B\Lambda$ and the relevant regressor covariance matrices. These are conditions for the stated probability limits, not empirical facts about an exchange.

Define

$$L = I_N - B\Lambda, \qquad H = L^{-1}, \qquad D = B\Gamma + \Delta_f, \tag{3}$$
$$P = HD, \qquad U = HB, \qquad V = H, \tag{4}$$

so that substitution gives the flow reduced form

$$q_t = P f_t + U u_t + V v_t, \tag{5}$$

with $\Sigma_{qq} = P\Sigma_f P^\top + U\Sigma_u U^\top + V\Sigma_v V^\top$.

Theorem 1 (Pseudo-true cross-impact matrices)**.** *Under the conditions above, the population coefficient in the regression of r_t on q_t is*

$$\operatorname{plim} \widehat{\Lambda}_{\mathrm{OLS}} = \Lambda + \underbrace{\Gamma \Sigma_f P^\top \Sigma_{qq}^{-1}}_{\textit{factor confounding}} + \underbrace{\Sigma_u U^\top \Sigma_{qq}^{-1}}_{\textit{simultaneity}}. \tag{6}$$

If the regression additionally controls for $h_t = f_t + \epsilon_t$, with ϵ_t uncorrelated with the system shocks, then with $R_f = \Sigma_f - \Sigma_f(\Sigma_f + \Sigma_\epsilon)^{-1}\Sigma_f$ and $Q_h = PR_fP^\top + U\Sigma_u U^\top + V\Sigma_v V^\top$,

$$\operatorname{plim} \widehat{\Lambda}_{q|h} = \Lambda + \Gamma R_f P^\top Q_h^{-1} + \Sigma_u U^\top Q_h^{-1}. \tag{7}$$

The proof is in Appendix A. Write

$$G := \operatorname{plim} \widehat{\Lambda}_{\mathrm{OLS}} - \Lambda \tag{8}$$

for the *confounding gap*. Everything that follows concerns its structure.

Corollary 1 (Spurious cross coefficients)**.** *Setting a structural entry $\Lambda_{ij} = 0$ does not imply that the population regression coefficient at (i, j) is zero. Either term in Eq. (6) can be off-diagonal, including when all structural off-diagonals vanish.*

Proposition 1 (Proxy precision is not elementwise bias monotonicity)**.** *Let Σ_f and two proxy-noise covariances be positive definite with $\Sigma_{\epsilon,1} \preceq \Sigma_{\epsilon,2}$, giving $R_{f,1} \preceq R_{f,2}$. The entries of $\operatorname{plim} \widehat{\Lambda}_{q|h} - \Lambda$ need not be ordered in absolute value, because Q_h^{-1} changes with proxy precision and reweights both bias components.*

Proposition 1 warns against scalar attenuation intuition. A better proxy removes residual factor variation in positive-semidefinite order, but a matrix coefficient is a ratio of covariance objects.

4 The Confounding Gap Is Low Rank

Theorem 2 (Rank of the confounding gap)**.** *Under the conditions of Theorem 1,*

$$\operatorname{rank}(G) \ \le \ K + \operatorname{rank}(B). \tag{9}$$

If $B = 0$ then $G = \Gamma\Sigma_f \Delta_f^\top \Sigma_{qq}^{-1}$ and $\operatorname{rank}(G) \le \min\{K, \operatorname{rank}(\Gamma), \operatorname{rank}(\Delta_f)\}$.

Proof. The first summand of Eq. (6) is $\Gamma \cdot (\Sigma_f P^\top \Sigma_{qq}^{-1})$, a product of an $N \times K$ and a $K \times N$ matrix, so its rank is at most K. The second is $\Sigma_u B^\top H^\top \Sigma_{qq}^{-1}$, whose rank is at most $\mathrm{rank}(B^\top) = \mathrm{rank}(B)$ because the remaining factors are nonsingular or cannot raise rank. Rank is subadditive over sums. When $B = 0$ the second summand vanishes and $P = \Delta_f$, giving the refinement. □

Equation (9) is the paper's organising result. It says the contamination in a cross-impact regression is not an arbitrary matrix: it lives in a space of dimension governed by the number of latent factors and the rank of same-bin feedback.

Corollary 2 (Spurious cross-impact is confined to a low-rank set)**.** *If $\Lambda = D$ is diagonal and $B = 0$, then* $\mathrm{plim}\,\widehat{\Lambda}_{\mathrm{OLS}} = D + G$ *with* $\mathrm{rank}(G) \le K$*, so the population coefficient matrix lies in*

$$\mathcal{D}_K := \{\, D + R \,:\, D \text{ diagonal},\ \mathrm{rank}(R) \le K \,\}. \tag{10}$$

Remark 1 (Low rank is not small). Membership in $\mathcal{D}_K$ places no bound on magnitude. At our registered fixture with $N = 30$, $K = 3$, $B = 0$, and a strictly diagonal Λ whose entries are drawn from $[0.2, 0.4]$, the induced spurious off-diagonal entries reach 0.2207 in absolute value, against realised own-impact terms spanning 0.2061 to 0.3953. The spurious cross-impact matrix is dense, of the same order as genuine own-impact, and entirely an artifact.

Remark 2 (What a factor control actually does). By Theorem 1, controlling for a noisy proxy replaces Σ_f with R_f and Σ_{qq} with Q_h. The controlled estimate is therefore another point of the form $\Lambda + \Gamma' W'^\top$: the control moves the estimate *along* the confounding directions rather than removing them. It identifies Λ only in the degenerate case $R_f = 0$.

5 Partial Identification

Fix $B = 0$, matching the registered empirical geometry, and normalise $\Sigma_f = I_K$ by absorbing scale into Γ and Δ_f. The observables are Σ_{qq}, Σ_{rr}, and $A := \Sigma_{rq}\Sigma_{qq}^{-1}$. With $W := \Sigma_{qq}^{-1}\Delta_f$ we have $G = \Gamma W^\top$ and

$$\Lambda = A - \Gamma W^\top. \tag{11}$$

Proposition 2 (Identified set)**.** *Given $(A, \Sigma_{qq}, \Sigma_{rr})$ and a factor count K, the identified set for Λ is*

$$\mathcal{I} = \left\{ A - \Gamma W^\top \;\middle|\; \begin{array}{l} W = \Sigma_{qq}^{-1}\Delta_f, \\ \Sigma_{qq} - \Delta_f \Delta_f^\top \succeq 0, \\ \Sigma_u \succeq 0 \end{array} \right\} \tag{12}$$

with $\Sigma_u = \Sigma_{rr} - \Lambda\Sigma_{qq}\Lambda^\top - \Lambda\Delta_f\Gamma^\top - \Gamma\Delta_f^\top\Lambda^\top - \Gamma\Gamma^\top$*. Every element reproduces the observed second moments exactly, so Λ is set-identified rather than point-identified.*

The proof is in Appendix B. The choice $\Gamma = 0$ is always feasible and returns $\Lambda = A$, the naive reading; the scientific question is how far $\mathcal{I}$ extends from it.

Table 2: Sharp identified intervals at the source-matched calibration ($N = 30$, $s_q = 0.2827$, $s_r = 0.32$, $d = 0.29$). A conditional analytic exhibit, not an estimate of any market's impact matrix.

o	A_{off}	Half-width	Identified interval
0.0029	0.01055	0.09431	$[-0.08375,\ 0.10486]$
0.0046	0.01225	0.09042	$[-0.07817,\ 0.10267]$

5.1 A closed-form sharp interval

Specialise to the permutation-invariant one-spike geometry used in our registered design: $K = 1$, $B = 0$, $m = \mathbf{1}_N/\sqrt{N}$, and

$$\Sigma_{qq} = q_0 I + (q_1 - q_0)mm^\top, \qquad q_1 = Ns_q, \tag{13}$$
$$\Sigma_{rr} = r_0 I + (r_1 - r_0)mm^\top, \qquad r_1 = Ns_r, \tag{14}$$

with $q_0 = (N - q_1)/(N - 1)$, $r_0 = (N - r_1)/(N - 1)$, $\Delta_f = h_q m$, $\Gamma = \gamma m$, and $h_q^2 = q_1 - q_0$. Since $\Sigma_{qq} m = q_1 m$,

$$G = \frac{\gamma h_q}{q_1} mm^\top, \qquad G_{ij} = \frac{\gamma h_q}{N q_1} \text{ for all } i, j. \tag{15}$$

The gap is a single constant added to every entry. Writing A_{diag} and A_{off} for the common entries of A, $t := NG_{ij}$, and $a_1 = A_{\mathrm{diag}} + (N-1)A_{\mathrm{off}}$ for the leading eigenvalue of A:

Proposition 3 (Sharp identified interval)**.** *The positive-semidefiniteness constraints reduce to the single inequality*

$$t^2 \le T^2, \qquad T^2 = \frac{(r_1 - q_1 a_1^2)(q_1 - q_0)}{q_1 q_0}, \tag{16}$$

feasible if and only if $r_1 \ge q_1 a_1^2$, and the identified set for the structural off-diagonal is exactly

$$\Lambda_{\mathrm{off}} \in \left[A_{\mathrm{off}} - \tfrac{T}{N},\ A_{\mathrm{off}} + \tfrac{T}{N} \right]. \tag{17}$$

The proof is in Appendix C. A bisection over the exact positive-semidefiniteness frontier reproduces T to a relative error below 10^{-10}.

Corollary 3 (Sign non-identification)**.** *At the calibration of Table 2 the identified interval contains zero at both registered structural endpoints. The identified half-width is* 7.4 *to* 8.9 *times the observed off-diagonal coefficient. Under the stated conventions the structural off-diagonal is therefore not identified even in sign from second moments.*

Figure 1 traces the interval across leading flow-commonality shares. The observed coefficient is visually indistinguishable from zero at this scale, while the identified set is wide throughout and contains zero everywhere on the grid.

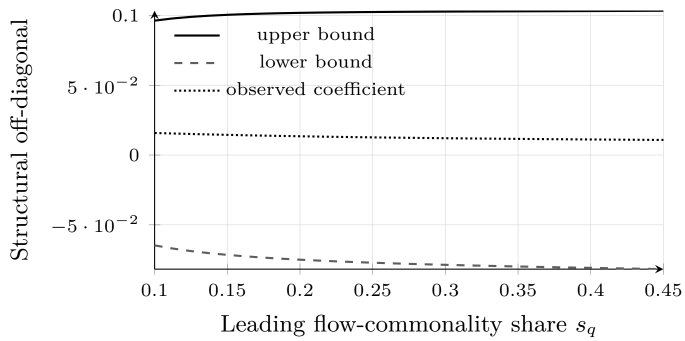

Figure 1: Sharp identified interval for the structural off-diagonal as leading flow commonality varies, at $s_r = 0.32$ and $d = 0.29$. The set contains zero across the whole grid, and the observed coefficient sits far inside it. A conditional analytic exhibit under the declared one-spike and isotropic-residual conventions, not a market estimate.

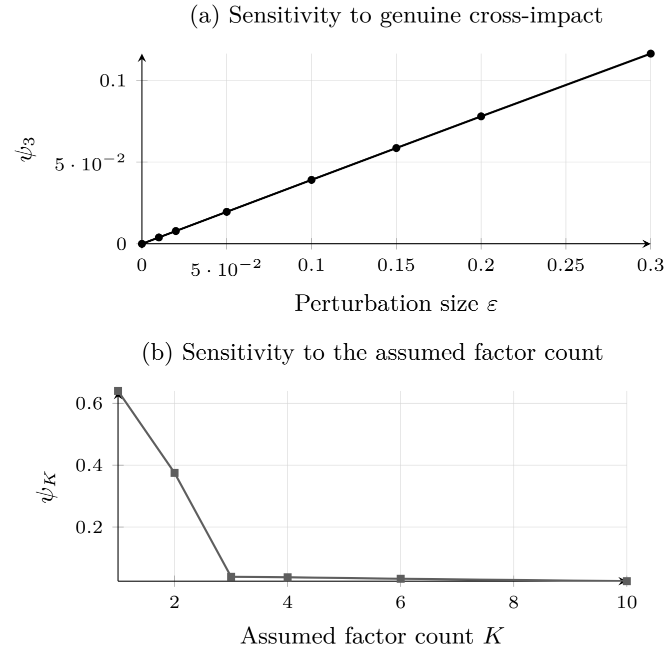

Figure 2: The diagnostic under both failure directions. Panel (a): ψ_3 is zero at exact diagonal-plus-rank-3 structure and strictly increasing in a genuine off-diagonal perturbation. Panel (b): the statistic is inflated when the factor count is understated and decays when it is overstated, with an elbow at the true value $K = 3$. Deterministic evaluations at test seed 1729.

6 A Refutable Restriction

Corollary 2 confines a purely confounded coefficient matrix to $\mathcal{D}_K$, which is a proper subset of $\mathbb{R}^{N\times N}$ whenever $K < N - 1$. Membership is therefore refutable.

Definition 1 (Low-rank departure statistic)**.** *For a coefficient matrix $\widehat{A}$ and factor count K, define*

$$\psi_K(\widehat{A}) = \frac{\min_{D,\,\mathrm{rank}(R)\le K} \big\|\widehat{A} - D - R\big\|_F}{\big\|\widehat{A} - \mathrm{diag}(\widehat{A})\big\|_F}, \tag{18}$$

the minimum over diagonal D and rank-K R, normalised by total off-diagonal energy.

The normalisation makes ψ_K scale-free and invariant to simultaneous row-and-column permutation of the assets, both verified numerically to 10^{-12}.

Proposition 4 (Population zero)**.** *Under H_0: Λ diagonal, $B = 0$, and exactly K latent factors, $\psi_K(\mathrm{plim}\,\widehat{\Lambda}_{\mathrm{OLS}}) = 0$.*

Proof. Immediate from Corollary 2: the minimand attains zero at $D = \Lambda$ and $R = G$. □

Direction of the test. A materially nonzero ψ_K is evidence *against* pure confounding and therefore *for* genuine structural cross-impact. This is the reverse of the conventional reading, under which a large off-diagonal coefficient is itself the evidence. Under our results off-diagonal magnitude carries no such information; only departure from $\mathcal{D}_K$ does.

Computation and conservatism. The minimisation in Eq. (18) is biconvex and is solved by alternating projection: with R fixed the optimal D is $\mathrm{diag}(\widehat{A} - R)$, and with D fixed the optimal R is the rank-K truncated singular value decomposition of $\widehat{A} - D$. Each half-step is the exact Frobenius minimiser over its block, so the objective is nonincreasing and bounded below. The limit is a stationary point rather than a certified global minimum, so the computed ψ_K is an *upper bound* on the true distance and the test is conservative toward failing to reject H_0. The alternating projection itself is standard [12]; the contribution here is the economic restriction it tests.

Two caveats stated up front. First, $\psi_K = 0$ does not prove Λ is diagonal: a structural matrix that is itself diagonal-plus-rank-K is observationally indistinguishable from a contaminated diagonal one. The test refutes; it does not confirm. Second, ψ_K depends on the assumed factor count. Overstating K shrinks the statistic mechanically and weakens the test; understating it inflates the statistic and can produce spurious rejection. Any application must report a sweep over K.

Figure 2 shows both behaviours. Panel (a) confirms strict monotonicity in the size of a genuine structural perturbation. Panel (b) shows the factor-count sensitivity, with a pronounced elbow at the true $K = 3$: the statistic falls from 0.6396 at $K = 1$ and 0.3748 at $K = 2$ to 0.0391 at $K = 3$, then decays slowly. The elbow is a practical way to read the factor count off the coefficient matrix itself.

Table 3: Cost error in the one-spike geometry, where the confounding direction is the equal-weight direction. A conditional analytic exhibit, not a market estimate.

Trade	True cost	Error	Relative
Equal-weight index	0.4234	+0.2296	**+54.23%**
Random unit vector	0.2950	+0.0160	+5.42%
Dollar-neutral pair	0.2854	0	**0.00%**

7 What It Costs to Be Wrong

Identification failure is only interesting if it costs something. A desk does not consume the impact matrix; it consumes the cost of a trade computed through it. Let a desk execute $x \in \mathbb{R}^N$ and pay expected impact cost $C(x, M) = x^\top Mx$.

Theorem 3 (Rank of the cost error)**.** *For any trade x, $C(x,A)-C(x,\Lambda) = x^\top Gx$. Since $\operatorname{rank}(G) \le K+\operatorname{rank}(B)$ by Theorem 2, the error vanishes identically on a subspace of trade space of dimension at least $N-K-\operatorname{rank}(B)$.*

Corollary 4 (Most trade directions are safe)**.** *At most $K + \operatorname{rank}(B)$ of the N directions in trade space are mispriced. The magnitude of the spurious off-diagonals is irrelevant to this count.*

This is the practical content of the rank bound. A dense, badly contaminated cross-impact matrix still misprices only a K-dimensional set of trades. In our fixture with $N = 30$, $K = 3$, $B = 0$ the immune subspace has dimension exactly 27, and a trade drawn from it has cost error 5.2×10^{-17}.

Which K directions are unsafe depends on the geometry. In a fixture with generic loadings the equal-weight basket is not special and bears about the same error as a random trade, −9.81% against −9.77%. In the permutation-invariant one-spike geometry, where the confounding direction *is* the equal-weight direction, Eq. (15) gives

$$C(x,A) - C(x,\Lambda) = g\,(\mathbf{1}^\top x)^2, \tag{19}$$

so the error is exactly proportional to the squared factor exposure of the trade. Table 3 reports the consequence.

An index basket is mispriced by more than half its true cost. A dollar-neutral pair is mispriced by exactly nothing. The operative question for a practitioner is therefore not whether the cross-impact matrix is identified, but whether the trade loads on the confounding direction.

8 Executing Under Partial Identification

Proposition 2 says the desk cannot learn Λ. It can still bound what a trade costs.

Proposition 5 (Identified cost interval)**.** *In the one-spike geometry every admissible structural matrix has the form $\Lambda = A - (t/N)\mathbf{1}\mathbf{1}^\top$ with $|t| \le T$, so*

$$C(x,\Lambda) \in \left[\, C(x,A) - \tfrac{T}{N}(\mathbf{1}^\top x)^2,\ C(x,A) + \tfrac{T}{N}(\mathbf{1}^\top x)^2 \,\right], \tag{20}$$

and the interval is sharp.

The half-width is again governed entirely by factor exposure, which yields the most useful consequence in the paper:

Corollary 5 (Cost can be identified where the matrix is not)**.** *In the one-spike geometry, a dollar-neutral trade has a degenerate cost interval. Its execution cost is point-identified even though the impact matrix producing it is set-identified with an interval containing zero.*

An unidentified parameter does not imply an unidentified cost. The scope qualifier is essential and easy to lose: dollar-neutrality delivers immunity only because the one-spike confounding direction *is* the equal-weight direction. In a general geometry the two come apart. At our fixture with generic loadings, a trade with $\mathbf{1}^\top x = 0$ to machine precision still carries a cost error of −18.24% of its true cost, because it is not orthogonal to the confounding subspace. The condition that actually confers immunity is Theorem 3's, membership of the null space of G; the one-spike geometry merely makes that null space easy to name.

For a desk that must take factor exposure, worst-case cost is $\widehat{C}(x) = x^\top A_s x + \frac{T}{N}(\mathbf{1}^\top x)^2$, convex whenever $A_s \succeq 0$. Minimising it subject to a target $c^\top x = q$ gives

Proposition 6 (Minimax-cost schedule)**.** *$x^*(\pi) = \frac{1}{2}\lambda M(\pi)^{-1}c$ with $M(\pi) = A_s + \pi\mathbf{1}\mathbf{1}^\top$ and $\lambda = 2q/(c^\top M(\pi)^{-1}c)$, evaluated at $\pi = T/N$. Its factor exposure is weakly smaller than the naive schedule's.*

Corollary 6 (When robustness is worthless)**.** *If the constraint pins the factor exposure, $x^*(\pi) = x^*(0)$ for every π. This covers a fixed total-quantity constraint $\mathbf{1}^\top x = Q$, an index-like target $c \propto \mathbf{1}$, and any target whose unconstrained optimum is already neutral.*

Corollary 6 is stated because a desk needs to know when not to bother. Table 4 confirms it: robustness earns its keep precisely when the desk retains discretion over factor exposure, and there it cuts exposure by roughly eighty percent. The closed form matches a 20,000-point grid search over feasible schedules exactly.

This is a static one-period impact model. It carries no decay kernel, risk aversion, or timing risk, and supports no profitability claim. The rank structure of the error does not depend on the schedule, so the immune-subspace conclusion is expected to survive in a multi-period quadratic-cost model, but that extension is not proved here.

Table 4: Minimax-cost schedule against the naive schedule. Both degenerate cases were predicted before implementation.

Target	Improvement	Exposure $\mathbf{1}^\top x$	
		naive	robust
Index-like, $c = \mathbf{1}$	0.00%	+1.000	+1.000
Neutral, $\mathbf{1}^\top c = 0$	0.00%	0.000	0.000
General	**3.11%**	−0.0663	**−0.0130**

Table 5: Realised size at nominal 5% and power at $T = 5000$, $N = 30$, $K = 3$, $B = 199$. Monte Carlo standard errors 0.018 and 0.022.

T	T/N^2	Plug-in size	Inflated
500	0.56	0.267	0.000
1,000	1.11	0.127	0.000
2,000	2.22	0.100	0.000
5,000	5.56	**0.040**	0.000
Power at $T = 5000$ *against perturbation size* ε			
$\varepsilon = 0.05$			0.070
$\varepsilon = 0.10$			0.270
$\varepsilon = 0.20$			**0.870**
$\varepsilon = 0.40$			1.000

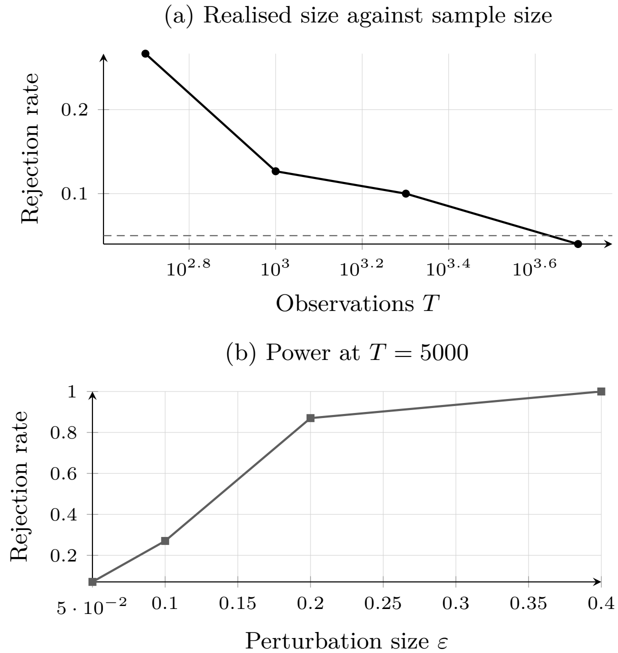

Figure 3: The departure test is valid only for large samples. Panel (a): realised size against the nominal 5% dashed line; the test over-rejects severely below roughly $T = 5N^2$ and is accurate at $T = 5000$. Panel (b): power against a dense off-diagonal perturbation at $T = 5000$. Monte Carlo standard errors 0.018 and 0.022.

8.1 A finite-sample null distribution

Proposition 4 is a population statement, which leaves "materially nonzero" undefined. We close that gap with a parametric plug-in bootstrap: refit the null projection $\widehat{A}_0$ from the estimate, redraw the estimation error from the ordinary-least-squares sampling law $\sqrt{T}(\widehat{A}_{i\cdot} - A_{i\cdot}) \to_d N(0, \sigma_i^2 \Sigma_{qq}^{-1})$, and take the upper $1 - \alpha$ quantile of $\psi_K(\widehat{A}_0 + E^*)$ as the critical value. The factor count is selected by an eigenvalue-ratio rule fixed before use, so it cannot be chosen after seeing a rejection.

The null manifold has dimension $N + K(2N - K) = 201$ against $N^2 = 900$ free parameters, so the restriction has codimension 699. Since $\psi_K(\widehat{A}) = O_p(T^{-1/2})$ under H_0, the test is consistent against any fixed alternative outside $\mathcal{D}_K$.

Table 5 reports a confirmatory study at a seed fixed after the design was registered. **The test controls size only for large samples.**

At $T = 5000$, roughly $5N^2$ observations, realised size is 0.040 against nominal 0.05 and power against a Frobenius-0.20 alternative is 0.870. Below that the test over-rejects badly, reaching 0.267 at $T = 500$. We state the usable range rather than implying general validity: at $N = 30$ the test should not be used below about $T = 5N^2$.

The cause is diagnosed rather than left open. The plug-in null is refitted to the noisy estimate, so the low-rank fit absorbs part of the estimation error and the bootstrap statistics are centred too low. We report a failed remedy alongside the working method: inflating the bootstrap error scale by the degrees-of-freedom factor $\sqrt{N^2/(N^2 - 201)} = 1.1347$ drives realised size to exactly zero at every sample size and removes all power, because the bias is in the centre of the null distribution and not its scale. A centre correction such as a double bootstrap would require its own design and is not attempted.

9 Known-Truth Verification

The probability limits were frozen before simulation code and random-number access, then evaluated along two algebraically independent population paths and tested in a single registered draw with $N = 30$, $K = 3$, and $T = 10{,}000{,}000$ independent Gaussian observations. The fixture is deliberately nontrivial, with $\rho(B\Lambda) = 0.4104$ and flow-covariance condition number approximately 348.1.

The gate statistic was the maximum elementwise relative discrepancy across both coefficient matrices, with no denominator floor, element exclusion, averaging substitute, or seed retry; passage required $D_{\text{gate}} < 10^{-3}$.

Table 6 reports the outcome. The sole frozen draw passed, and all 1,800 target coefficients lay inside named 95% family-wise classical homoskedastic Student-t Bonferroni intervals. Immediate replay reused all validated checkpoints and reproduced the summary, estimates, and success marker byte for byte.

The lower panels report the new checks. The rank

Table 6: Completed known-truth verification and the new deterministic checks.

Quantity	Verified value
Registered $T = 10^7$ experiment	
Assets / factors / observations	$30/3/10^7$
Reported coefficients	1,800
OLS max relative discrepancy	5.6395×10^{-4}
Proxy-control max discrepancy	5.1237×10^{-4}
Gate threshold	1.0×10^{-3}
Targets inside intervals	1,800 / 1,800
Rank bound, $\mathrm{rank}(G)$ *observed vs.* $K + \mathrm{rank}(B)$	
$\mathrm{rank}(B) = 0$	3 vs. 3
$\mathrm{rank}(B) = 1$	4 vs. 4
$\mathrm{rank}(B) = 2$	5 vs. 5
$\mathrm{rank}(B) = 30$	30 vs. 33
Diagnostic	
ψ_3 at exact structure	0, below a 10^{-12} floor
Permutation-invariance residual	$< 10^{-12}$
One-spike gap constancy	$< 10^{-12}$
Closed form vs. bisection	$< 10^{-10}$

bound binds exactly at every arm where it can bind and is never violated; at $\mathrm{rank}(B) = 30$ the observed rank is capped by the asset count. The diagnostic attains its predicted population zero and both invariances. Every entry in these two panels is a deterministic algebraic evaluation at test seed 1729, so no interval method applies and no stochastic comparison is performed.

What this establishes is narrow. The symbolic expressions and the primitive covariance implementation agree; row-stacked software returns the expected transpose; and the streamed sufficient-statistic path recovers the registered target at the intended dimension. It does not establish identification, empirical fit, causal timing, interval coverage under market dependence, or economic usefulness.

10 Confronting Published Evidence

Equation (15) yields a prediction that can be checked against already-published summary statistics without touching market data. In the permutation-invariant geometry a single factor control shifts *every* cross-coefficient by one common constant. Two implications follow, and they can disagree.

Dispersion should be invariant. A constant shift moves the cross-sectional mean and leaves the cross-sectional standard deviation unchanged. Capponi and Cont [9] report a mean cross coefficient moving from 0.032 to -0.039, a shift of -0.071, while the cross-sectional standard deviation stays at 0.06; the own-coefficient standard deviation moves only from 0.78 to 0.77. Both changes are at or within the reported two-decimal precision. This is the predicted signature, and it is unit-free, so it requires none of the variable standard deviations the source does not report.

Distribution shape should be invariant too, and here the check fails. Under an exactly constant shift the post-control negative fraction should equal the pre-control mass lying below the shift magnitude. A Gaussian approximation to the reported mean and standard deviation puts that at 0.7422, against a reported 0.8446 – a gap of 0.1024, exceeding our declared tolerance of 0.05.

We report the failure rather than remove it. The registered protocol forbids retuning the one-spike convention to close such a gap. The most likely explanation is loading heterogeneity beyond one common factor, which the one-spike convention assumes away and which the Gaussian approximation to an empirically skewed cross-section compounds. Note also that the discrepancy does not bear on Theorem 2, which is an inequality in K and is not weakened by the presence of additional factors; it bears on the sharper constant-shift specialisation of Eq. (15).

Both checks are conditional analytic exhibits at published summary statistics. Neither is an estimate of any market's impact matrix, and the published dispersion figures are not confidence intervals and are not treated as such.

11 Registered Falsification Protocol

The empirical stage is governed by a protocol registered before any data is opened. Its question is deliberately narrower than identification:

> Can confounding alone be materially large in a transparent model whose observable flow commonality and factor alignment match opened primary-source summaries, even when the estimator receives a favourable factor proxy?

The design sets $B = 0$, so a positive result cannot be explained by price chasing. It retains $N = 30$, uses one permutation-invariant factor, sets proxy reliability to 0.95, and evaluates 17 frozen homogeneous structural off-diagonal values from 0.0029 to 0.0046. Three smooth estimators bind at every grid point: an observable integrated-top-ten-OFI proxy-control condition-ridge estimator, an oracle-flow condition-ridge estimator, and an oracle-flow homogeneous three-slope OLS. The first is the binding observable opponent; the two oracles remove measurement error and high-dimensional inversion as alternative explanations.

A reconstruction of six published specifications supplies a separate veto. Because the source paper does not disclose feature scaling, fold chronology, penalty grid, tie rule, or solver tolerance, the executable version is a preregistered protocol reconstruction rather than author code, and every omitted numerical choice is frozen in advance.

For focal estimate $\widehat{\Lambda}_{01}$, truth o, and a standard error from 499 shared whole-date bootstrap replicates, passage requires

$$\left|\widehat{\Lambda}_{01} - o\right| - 0.50|o| > 3\,SE_{\text{boot}}. \tag{21}$$

Before the research draw can exist, 100 independent validation superpanels must license the exact final procedure through a null-grid size condition and a power condition with named Clopper–Pearson and Wilson bounds.

The interpretation of every branch is fixed in advance. If all binding candidates pass, the result supports material confounding within the registered source-matched model class only. If any binding candidate fails, the protocol does not pass, and no diagnostic can rescue it. If size, power, numerical, provenance, or resource admission fails, the research draw stays sealed as a design failure rather than a market null. A premise-killing null requires a separately registered sharp upper bound over the source-compatible class; a single negative fixture is insufficient.

This protocol is specified here, not reported. No registered stochastic stream has run.

12 Limitations

The completed evidence is narrow by construction. The known-truth fixture is Gaussian and large-sample, its zero contemporaneous shock cross-covariances are substantive assumptions, and its dense positive targets do not reproduce the small, sign-sensitive coefficients central to the empirical debate. Interval inclusion there is one realisation, not an estimated coverage probability.

The new results carry their own limits. Theorem 2 is an inequality, so a high observed rank is uninformative if K is misspecified, and the rank of the gap is not separately observable from the rank of a genuinely low-rank Λ. Proposition 2 assumes $B = 0$; with feedback the identified set is larger, not smaller, so Corollary 3 is conservative in that direction only. The closed form of Proposition 3 is conditional on the one-spike and isotropic-residual conventions, which are declared maximum-entropy choices and not identified features of any exchange. The ψ_K statistic is an upper bound from a stationary point, has no derived sampling distribution here, and cannot confirm diagonality.

The published-evidence exercise of Section 10 uses summary statistics that carry no inferential intervals, so it supports consistency statements only and admits no p-value.

Finally, predictive performance and structural identification remain distinct. An estimator can forecast well while targeting the wrong impact matrix, and a structurally interpretable estimator can be economically useless after costs. Nothing here supports a trading rule, execution-cost saving, or deployment claim.

13 Conclusion

Off-diagonal return-on-flow coefficients do not identify structural cross-impact by notation alone, and the failure has a precise shape. The gap between the population coefficient and the structural matrix has rank at most $K + \operatorname{rank}(B)$. That single fact carries the paper: it explains why a diagonal truth can generate a dense cross-impact matrix of realistic magnitude, why the structural matrix is set-identified rather than point-identified, why a factor control moves an estimate along the contamination rather than out of it, and why the resulting ambiguity is wide enough at published commonality levels to leave the structural off-diagonal unsigned.

The same fact supplies the remedy. Low rank is refutable, and ψ_K measures departure from it with the polarity the problem requires: rejection is evidence for genuine cross-impact, not against it. Whether real markets reject is an empirical question this paper deliberately does not answer. It registers the protocol that will, and reports the theory that makes the answer interpretable when it arrives.

Declarations

Authorship. Mehmet Demir Güven is the sole author. He conceived the research question, fixed the design and the preregistered constraints under which it was carried out, directed the work, reviewed and accepted the results reported here, and takes full responsibility for the content of this paper, including any errors it contains.

Computational assistance. Generative AI tools were used during this project to assist with algebraic derivation, software implementation, and manuscript preparation. Every formal result was checked against independent numerical verification, every reported value regenerates from committed deterministic code, and the preregistration, derivations, and decision ledgers are public so that any claim can be traced to the evidence behind it. Responsibility for the correctness of the work rests with the author. No AI system is an author of this paper.

Funding and institutional status. This independent research received no funding from ETH Zürich. The affiliation records the author's status as a computer science student only. ETH Zürich did not sponsor, approve, or endorse the research, and the paper does not represent the university's views.

Data and code availability. No external market data was accessed. Every numerical value reported here is regenerated by committed, deterministic code with a single command; the repository, preregistration, derivations, and ledgers are public.

Competing interests. The author declares none.

A Proof of Theorem 1

From Eqs. (1)–(2), $(I - B\Lambda)q_t = (B\Gamma + \Delta_f)f_t + Bu_t + v_t$, which yields Eq. (5). Mutual zero cross-covariances give $\mathrm{Cov}(f_t, q_t) = \Sigma_f P^\top$ and $\mathrm{Cov}(u_t, q_t) = \Sigma_u U^\top$. Using $r_t = \Lambda q_t + \Gamma f_t + u_t$,

$$\Sigma_{rq} = \Lambda\Sigma_{qq} + \Gamma\Sigma_f P^\top + \Sigma_u U^\top, \tag{22}$$

and right multiplication by Σ_{qq}^{-1} proves Eq. (6).

For the proxy-controlled regression, the linear-projection residual of f_t after controlling for $h_t = f_t + \epsilon_t$ is $f_t^\perp = f_t - \Sigma_f(\Sigma_f + \Sigma_\epsilon)^{-1}h_t$ with variance R_f. Frisch–Waugh–Lovell residualisation gives $q_t^\perp = Pf_t^\perp + Uu_t + Vv_t$ and $r_t^\perp = \Lambda q_t^\perp + \Gamma f_t^\perp + u_t$, so $\mathrm{Var}(q_t^\perp) = Q_h$ and $\mathrm{Cov}(r_t^\perp, q_t^\perp) = \Lambda Q_h + \Gamma R_f P^\top + \Sigma_u U^\top$. Right multiplication by Q_h^{-1} proves Eq. (7). □

B Proof of Proposition 2

Necessity. Any structural tuple consistent with the observables must satisfy Eq. (11) by Theorem 2 with $B = 0$, and its residual covariances must be positive semidefinite.

Sufficiency. Given (Γ, Δ_f) satisfying both semidefiniteness constraints, set $\Lambda = A - \Gamma W^\top$, $\Sigma_v = \Sigma_{qq} - \Delta_f\Delta_f^\top$, and Σ_u as stated. The resulting tuple is a valid instance of Eqs. (1)–(2) with $B = 0$, and reversing the algebra reproduces Σ_{qq}, Σ_{rr}, and Σ_{rq}. Hence every element of $\mathcal{I}$ is observationally equivalent. □

C Proof of Proposition 3

Decompose every matrix in the m direction and its orthogonal complement, on which all matrices act as multiples of the identity.

First, $\Sigma_v = \Sigma_{qq} - \Delta_f\Delta_f^\top = q_0 I \succeq 0$ automatically, imposing no restriction.

Next, $\Lambda m = \lambda_1 m$ with $\lambda_1 = d + (N-1)o$, and Λ acts as $(d-o)I$ on $m^\perp$. With $\Gamma = \gamma m$ and $\mathrm{Cov}(q_t, f_t) = h_q m$,

$$\Sigma_u = \Sigma_{rr} - \Lambda\Sigma_{qq}\Lambda^\top - (2\gamma h_q\lambda_1 + \gamma^2)mm^\top. \tag{23}$$

On $m^\perp$ the eigenvalue is $r_0 - (d-o)^2 q_0$; since $d - o = A_{\mathrm{diag}} - A_{\mathrm{off}}$, this is free of t and restricts nothing. On m the eigenvalue is $r_1 - \lambda_1^2 q_1 - 2\gamma h_q\lambda_1 - \gamma^2$, so $\Sigma_u \succeq 0$ requires

$$\gamma^2 + 2\lambda_1 h_q\gamma + \lambda_1^2 q_1 - r_1 \le 0. \tag{24}$$

Substituting $\gamma = tq_1/h_q$ and $\lambda_1 = a_1 - t$, the middle terms combine as $q_1(a_1 - t)(a_1 + t) = q_1(a_1^2 - t^2)$, giving

$$\frac{t^2 q_1^2}{q_1 - q_0} + q_1 a_1^2 - q_1 t^2 \le r_1, \tag{25}$$

that is $t^2 q_1 q_0/(q_1 - q_0) \le r_1 - q_1 a_1^2$, which is Eq. (16). Since $\Lambda_{\mathrm{off}} = A_{\mathrm{off}} - t/N$, Eq. (17) follows. □

D Equivalent Primitive Forms

With $\Omega_q = D\Sigma_f D^\top + B\Sigma_u B^\top + \Sigma_v$ and $\Omega_{q|h} = DR_f D^\top + B\Sigma_u B^\top + \Sigma_v$, and using $\Sigma_{qq} = H\Omega_q H^\top$ and $Q_h = H\Omega_{q|h}H^\top$, Theorem 1 is equivalently

$$\operatorname{plim}\widehat{\Lambda}_{\mathrm{OLS}} = \Lambda + (\Gamma\Sigma_f D^\top + \Sigma_u B^\top)\Omega_q^{-1}(I - B\Lambda), \tag{26}$$
$$\operatorname{plim}\widehat{\Lambda}_{q|h} = \Lambda + (\Gamma R_f D^\top + \Sigma_u B^\top)\Omega_{q|h}^{-1}(I - B\Lambda). \tag{27}$$

These forms show the decomposition is determined entirely by structural primitives under the stated zero-cross-covariance assumptions, and make the rank bound of Theorem 2 visible directly in the primitives.

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